Word problems on the motion of two bodies

Towards each other, in the same direction, and the closing speed that decides both.

LESSON 129–131 | 3 × 40 MIN | MATEMATIKA 6, §28 | S7 12 RATES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Towards each other	25 min
The place of meeting	30 min
Overtaking	35 min
Head starts	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the closing speed for two bodies approaching.
- Use the difference of speeds for one overtaking another.
- Find the time and place of meeting.
- Draw a diagram before writing any arithmetic.

TEXTBOOK

National	pp. 379–388
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.9

01 Explanation

Towards each other

Two bodies approaching close the gap at the **sum** of their speeds.

$$\text{time to meet} = \frac{\text{distance apart}}{v_1 + v_2}$$

APART	SPEEDS	CLOSING SPEED	TIME TO MEET
300 km	*60* and *40*	*100 km/h*	*3 h*
240 km	*70* and *50*	*120 km/h*	*2 h*
12 km	*5* and *3*	*8 km/h*	*1.5 h*

HERE THE SPEEDS REALLY DO ADD

Not because speeds can be added in general, but because each hour the gap shrinks by both distances at once. The sum is the speed of the *gap*, not of either body.

The place of meeting

STEP	FOR *300* KM APART AT *60* AND *40*
closing speed	*100 km/h*
time to meet	*3 h*
first travels	*180 km*
second travels	*120 km*
check	*180 + 120 = 300* ✓

THE FASTER ONE COVERS MORE OF THE GAP

In the same time, distances are in the ratio of the speeds — here *60 : 40 = 3 : 2*, so the *300* km splits as *180* and *120*.

Overtaking

Two bodies going the same way close the gap at the **difference** of their speeds.

$$\text{time to catch} = \frac{\text{head start}}{v_1 - v_2}$$

HEAD START	SPEEDS	DIFFERENCE	TIME TO CATCH
30 km	*80* and *60*	*20 km/h*	*1.5 h*
12 km	*15* and *12*	*3 km/h*	*4 h*
100 m	*8* and *6* m/s	*2 m/s*	*50 s*

IF THE SPEEDS ARE EQUAL, THE GAP NEVER CLOSES

A difference of zero means no catching up, however long the chase. The formula says so by dividing by zero, which is the mathematics agreeing with common sense.

Head starts

PROBLEM	HEAD START	ANSWER
a cyclist at *12* leaves *1* h before a car at *60*	*12 km*	caught after *0.25* h
a walker at *5* leaves *2* h before a cyclist at *15*	*10 km*	caught after *1* h
a bus at *60* leaves *30* min before a car at *80*	*30 km*	caught after *1.5* h

TURN THE TIME HEAD START INTO A DISTANCE HEAD START

An hour's start at *12* km/h is a *12* km gap. Once the head start is a distance, the chase formula applies directly.

02 Worked examples

EXAMPLE 1

Two cars *300* km apart drive towards each other at *60* and *40* km/h. When do they meet, and where?

1. Closing speed: *100* km/h.
2. Time: $300 \div 100 = 3$ h.
3. First: *180* km; second: *120* km.

   Check: they total *300* ✓

ANSWER

After *3* h, *180* km from the first

EXAMPLE 2 **A bus leaves at 60 km/h. Half an hour later a car follows at 80. When does it catch up?**

1. Head start: $60 \cdot 0.5 = 30$ km.
2. Difference of speeds: 20 km/h.
3. $30 \div 20 = 1.5$ h after the car starts.

ANSWER

After 1.5 hours

EXAMPLE 3 **Two walkers 12 km apart walk towards each other at 5 and 3 km/h. When do they meet?**

1. Closing speed: 8 km/h.
2. $12 \div 8 = 1.5$ h.
3. They meet 7.5 km from the first.

   $5 \cdot 1.5.$

ANSWER

After 1.5 hours

03 Interactive model

Two pupils walking towards each other along the corridor, timed, is a five-minute experiment that makes the closing speed obvious.

Sum or difference?

Two bodies approaching close the gap at:

- the sum of the speeds
- the difference
- the average
- the faster speed

One overtaking another closes at:

- the sum
- the difference
- the average
- the slower speed

300 km apart at 60 and 40: they meet after:

- 3 h
- 5 h
- 7.5 h
- 30 h

The first car has then travelled:

- 120 km
- 150 km
- 180 km
- 300 km

A 30 km head start with speeds 80 and 60: caught after:

- 0.375 h
- 1.5 h
- 2 h
- 3 h

A bus at 60 leaving 30 min early has a head start of:

- 30 km
- 60 km
- 0.5 km
- 120 km

If both speeds are equal, the chaser:

- catches up slowly
- never catches up
- catches up at once
- falls behind

Two walkers 12 km apart at 5 and 3 meet after:

- 1.5 h
- 2 h
- 4 h
- 6 h

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To approach	Yaqinlashmoq	Сближаться
To meet	Uchrashmoq	Встретиться
Closing speed	Yaqinlashish tezligi	Скорость сближения
To overtake	Quvib yetmoq	Догонять
Same direction	Bir yo‘nalishda	В одном направлении
Opposite directions	Qarama-qarshi	Навстречу
Meeting point	Uchrashuv nuqtasi	Точка встречи
Head start	Oldindan boshlash	Фора

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Towards each other, the closing speed is:

- $v_1 + v_2$
- $v_1 - v_2$
- $v_1 v_2$
- v_1

In the same direction it is:

- $v_1 + v_2$
- $v_1 - v_2$
- the average
- the larger

240 km apart at 70 and 50: the time to meet is:

- 2 h
- 4 h
- 12 h
- 1.2 h

The distances covered are in the ratio of:

- the times
- the speeds
- the total
- nothing

A 12 km head start with speeds 15 and 12:

- 0.8 h
- 4 h
- 1 h
- 12 h

The first step in these problems is:

- a formula
- a diagram
- a calculation
- a guess

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *300* km apart at *60* and *40*: the closing speed

   ANSWER *100 km/h*

2. And the time to meet

   ANSWER *3 h*

3. *240* km apart at *70* and *50*

   ANSWER *2 h*

4. *12* km apart at *5* and *3*

   ANSWER *1.5 h*

5. A *30* km head start, speeds *80* and *60*

   ANSWER *1.5 h*

6. A *12* km head start, speeds *15* and *12*

   ANSWER *4 h*

7. Equal speeds in the same direction

   ANSWER The gap never closes

Medium · the standard question

7 PROBLEMS

1. Where do the *300* km cars meet?

   ANSWER *180* km from the first

2. A bus at *60* leaves *30* min before a car at *80*

   ANSWER Caught after *1.5 h*

3. A walker at *5* leaves *2* h before a cyclist at *15*

   ANSWER Caught after *1 h*

4. A cyclist at *12* leaves *1* h before a car at *60*

   ANSWER Caught after *0.25 h*

5. Runners at *8* and *6* m/s with a *100* m start

   ANSWER *50 s*

6. Two trains *420* km apart at *90* and *120*

   ANSWER *2 h*

7. Where do they meet?

   ANSWER *180* km from the first

Hard · stretch and reason

7 PROBLEMS

1. Two cars *450* km apart meet after *3* h; one does *80*: the other

   ANSWER *70 km/h*

2. A car catches a bus *40* km ahead in *2* h: the difference in speeds

   ANSWER *20 km/h*

3. Two walkers *9* km apart at *4* and *5*: where do they meet?

   ANSWER *4* km from the first

4. A boat and a raft *24* km apart moving towards each other at *10* and *2*

   ANSWER *2 h*

5. A train *200* m long passing a pole at *20* m/s

   ANSWER *10 s*

6. Why does the gap close at the sum when they approach?

   ANSWER Each covers part of it in the same time

7. Two cyclists start together at *12* and *18*: the gap after *2* h

   ANSWER *12* km

07 Homework

Homework — 5 tasks

Draw the two bodies and an arrow each before choosing sum or difference.

1. Two cars *420* km apart drive towards each other at *90* and *120* km/h. When do they meet?
2. How far has each travelled by then?
3. A bus at *50* km/h leaves *1* hour before a car at *75*. When does the car catch it?
4. Two runners *120* m apart run in the same direction at *7* and *5* m/s. When is the first caught?
5. Explain why the closing speed is the sum in one case and the difference in the other.